Multimodal Lung Nodule Malignancy Prediction & Subtype Discovery

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ABSTRACT

- Accurate lung nodule malignancy prediction is critical for early diagnosis.
- CT-based deep learning performs well but lacks clinical context.
- We propose a multimodal framework combining:
  - **3D CT imaging (3D ResNet-18)**
  - **Structured radiologist annotations modeled using an MLP**
- We combine both using a validation-optimized late fusion approach.
- Evaluated on the LIDC-IDRI dataset.
- Grad-CAM highlights clinically relevant regions such as irregular boundaries and texture variations
- Fused embeddings enable subtype discovery identifying three clinically meaningful risk groups.
- Results demonstrate the value of **multimodal learning** for accurate, interpretable lung cancer prediction.

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Introduction

- Lung cancer is one of the leading causes of cancer-related deaths worldwide.
- Early and accurate diagnosis is essential for improving patient outcomes.
- CT imaging plays a key role in detecting lung nodules.
- Deep learning models on CT scans show strong performance.
- However, they often miss clinically meaningful information from radiologist annotations.
- Structured approaches capture clinical knowledge but lack imaging context.
- These methods provide complementary strengths.
- This motivates combining them using multimodal learning.

Motivation & Contributions

Motivation:

- CT-based models capture spatial features but miss clinical semantics.
- Radiologist annotations encode expert knowledge but lack imaging context
- Single-modality approaches fail to capture complete diagnostic information.

Contributions:

- Multimodal framework combining 3D CT imaging and radiologist annotations
- Late fusion improves performance over CT-only and structured-only models
- Enables subtype discovery and interpretability using clustering and Grad-CAM.

Methodology

- **Dataset:** LIDC-IDRI dataset (1000 nodules) with malignancy labels (1–5) from radiologist annotations; ambiguous cases excluded.
- **CT Imaging Branch:** 3D CT patches (64×64×64) are processed using a 3D ResNet-18 to extract spatial and morphological features. The model outputs a malignancy probability (**p_CT**) based on learned imaging representations.
- **Structured Feature Branch:** Radiologist annotations are converted into numerical feature as input. Features include spiculation, texture, subtlety, calcification, and internal structure modeled using a multilayer perceptron (MLP), producing malignancy probability (**p_struct**) based on structured data.
- **Fusion Strategy:** Weighted late fusion combines both modalities.: $p = \alpha\ p\_CT + (1 - \alpha)\ p\_struct$, where $\alpha = 0.38$ is selected using validation data to balance imaging and semantic contributions.

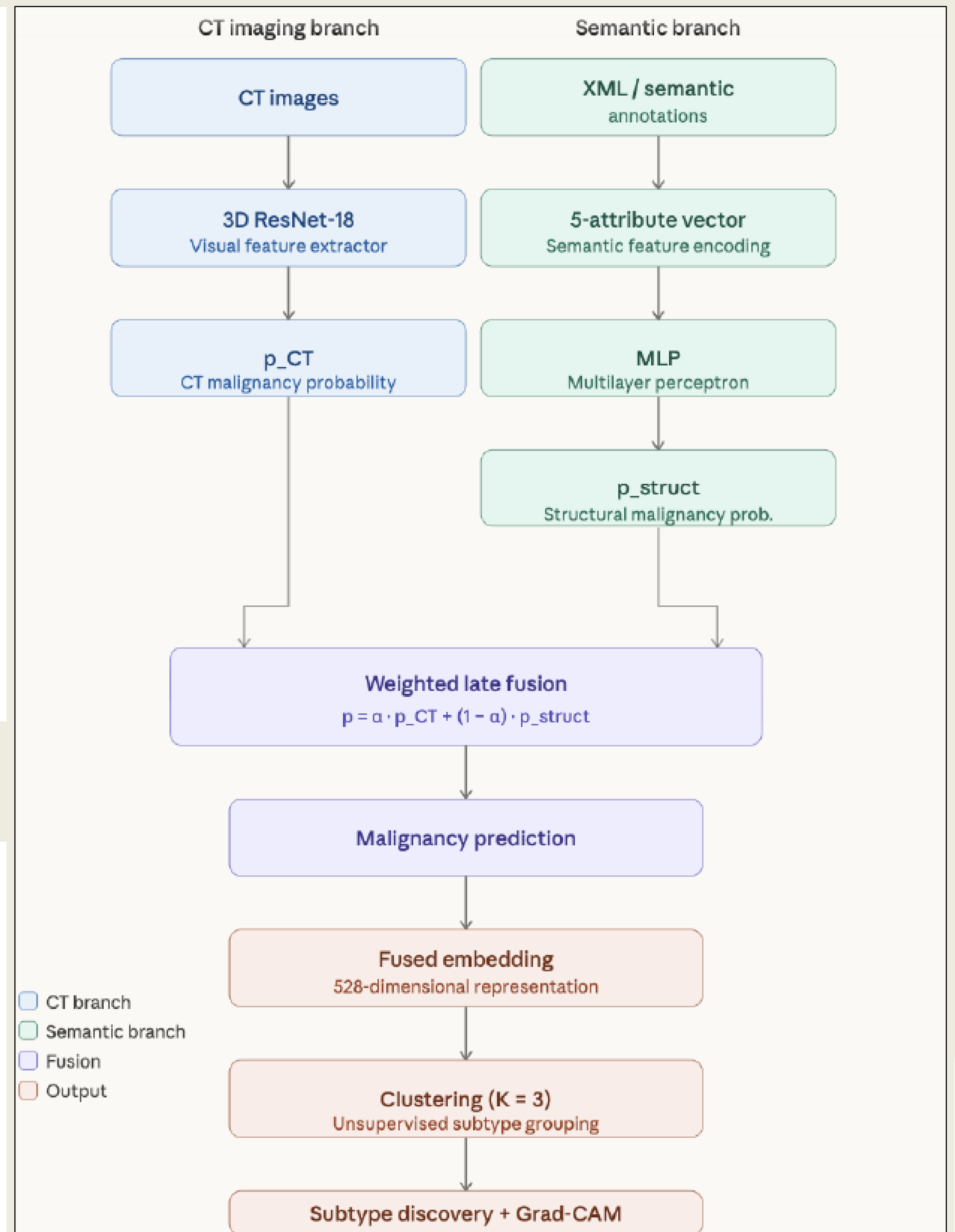


Figure 1. Proposed multimodal framework combining CT imaging and structured features via late fusion.

Embedding & Analysis:
The fused representation is further used to generate embeddings for malignant nodules. These embeddings are analyzed using clustering (K = 3) to identify distinct subtypes. Grad-CAM is applied to the CT branch to provide visual interpretability of model decisions

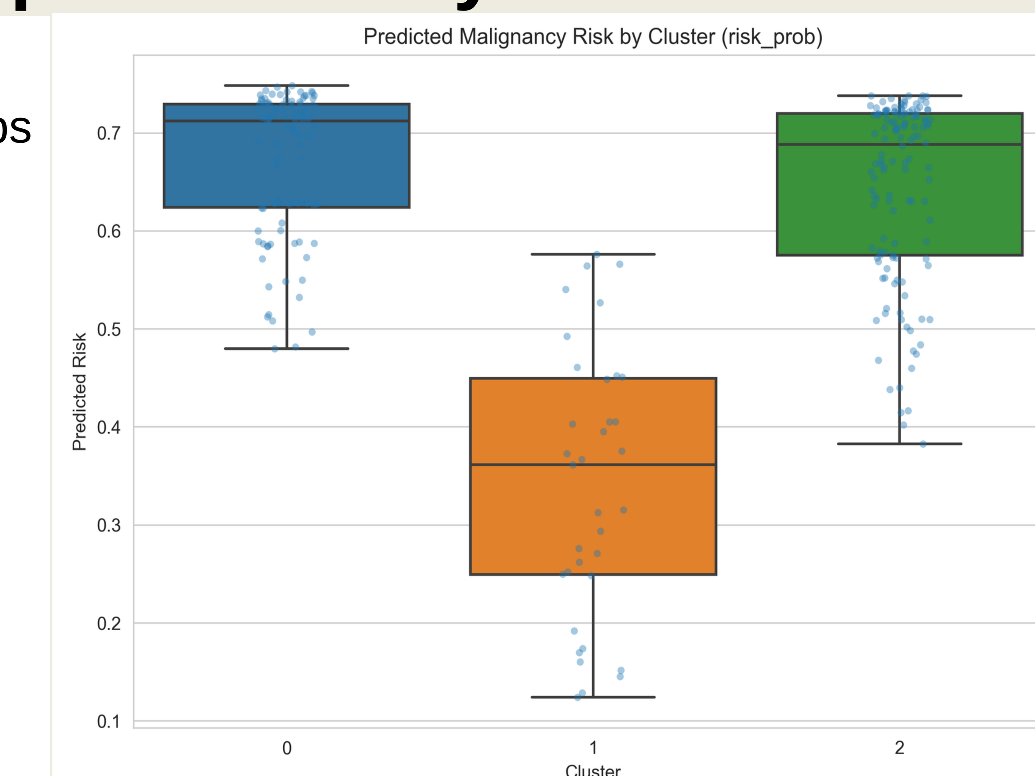
Results

MODEL	AUC	Precision	Recall	F1	Accuracy
CT-Only	0.9491	0.8333	0.8861	0.8589	0.9034
Struct-Only	0.9598	0.8481	0.8481	0.8481	0.8992
Late Fusion	**0.9747**	**0.9605**	**0.9241**	**0.9419**	**0.9622**

- Late fusion outperforms CT-only and structured-only models, achieving the highest **AUC (0.9747)**, **F1-score (0.9419)**, and **accuracy (0.9622)**.
- This shows that combining imaging and semantic features improves prediction performance.

Clustering & Subtype Discovery

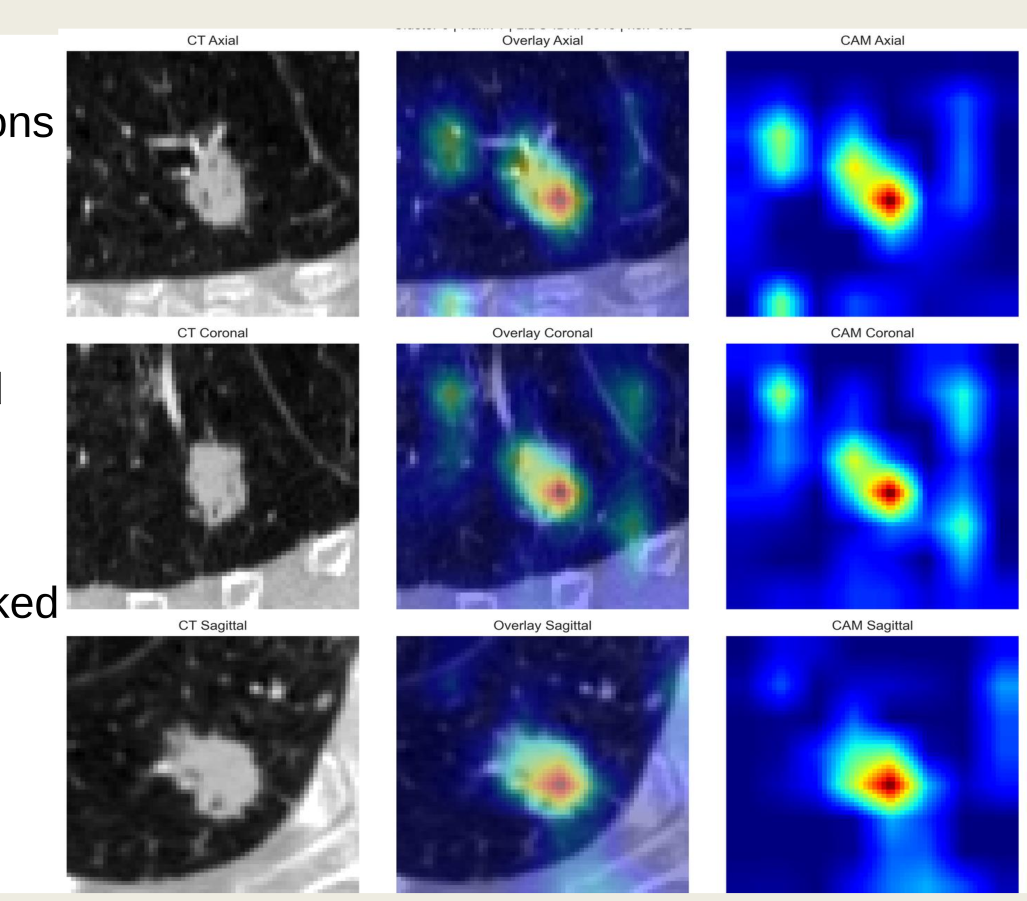
- KMeans identifies three clinically meaningful subgroups
- **Cluster 0 (high-risk):** Nodules with irregular boundaries and higher predicted risk
- **Cluster 1 (low-risk):** Nodules with smoother structures and lower predicted risk
- **Cluster 2 (intermediate-risk):** Nodules with mixed features and moderate risk
- Clusters align with radiologist annotations & risk scores.



- High-risk clusters (0, 2) show higher feature intensity, clinically associated with malignancy
- Low-risk cluster (1) shows lower feature intensity and lower predicted risk, less aggressive nodules
- High clustering stability (ARI ≈ 0.96) indicates robustness of the discovered subtypes
- The box plot shows clear risk differences, indicating the clusters represent meaningful clinical groups, not random ones.

Grad-CAM (Model Interpretability)

- Grad-CAM highlights regions within lung nodules driving model predictions
- **Cluster 0 (high-risk):** Focus on irregular margins and dense structures, linked to higher malignancy.
- **Cluster 1 (low-risk):** Weak, localized activations, linked to lower malignancy.
- **Cluster 2 (intermediate-risk):** Moderately distributed attention, linked to moderate malignancy.
- Model focuses on clinically relevant regions, with activations aligned to clinical features, supporting interpretability.



Conclusion & Future Work

- Multimodal fusion of CT imaging and radiologist annotations improves malignancy prediction and performance.
- The framework enables subtype discovery and provides interpretable, clinically useful insights.
- Future work includes validating on larger multi-institutional datasets and include ambiguous nodules for better generalization.

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